

Unsupervised Value Discovery: Locating the Value Structures of an Agent from Passive Observation by their Empowerment–Plasticity Asymmetry

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Abstract

Unsupervised agent discovery (UAD) locates agents in raw multivariate time series by finding variable sets that approximately satisfy a Markov-blanket condition against their surroundings. We ask a question one level inward: within a discovered agent, which parts are its *value structures*, the components of its informational structure that correspond to its goals, objectives, or desires? We propose that this role has a directional information-theoretic signature: values are the components that most strongly drive the world while being least driven by it, maximally empowered and minimally plastic. This inverts a familiar AI safety assumption that a system first develops values and then acts to make them empowered and stable; we hypothesize instead that whatever is empowered and stable thereby *serves as* the system’s values, and that this structural property is passively detectable. We operationalize the hypothesis, as a *value signature* on directed-information scores gated by procedure-mirroring noise floors, and evaluate it in three experiments, using worlds built on an existing UAD benchmark. These show the following. (i) In a single-agent world, the value signature identifies a planted goal, and only it, when using measurement conventions that can see through the world’s XOR-style composition. (ii) Across four more complex worlds, including an eight-agent, 49-variable colony world, only a block-level measurement convention, which fuses discovered agents and offers every block as a conditioning ‘key’, recovers the planted goals. (iii) Against adversarial impostors, a fast environment-driven ‘captured goal’ is refused, but a slow, history-triggered captured goal is wrongly identified by the value signature (a follow-up experiment showed this was due to its leak-tolerant intake clause rather than to any limit of passive observation: a strictly gated passive stability test refuses the same impostor), but a noiseless goal twin is passively indistinguishable in principle and can only be identified with an interventional yardstick generalizing the benchmark’s goal-flip operation, locating the limits of passive detection. We report limitations (single-variable values; lag-1 conditioning) and directions for further work.

1 Introduction

The standard route to an agent’s objectives runs through its behaviour: observe choices, infer the utility or reward that rationalizes them [4, 6]. This route has well-known difficulties (non-identifiability, dependence on rationality assumptions) and one that is rarely foregrounded: an agent

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may *conceal* its objectives, for instance by injecting noise into its decisions, so that the behavioural signal is deliberately weakened. This paper explores a complementary route. Rather than inferring values from what a system does, we look for its value structures in the *informational structure* of the system itself: which internal component most stably drives behaviour and impact? If such a component can be located structurally, it can be isolated and studied directly, recovering the signal that behavioural inference alone would lose.

Our starting point is unsupervised agent discovery (UAD) [8], which finds agents in raw time series as variable clusters that approximately satisfy a Markov-blanket condition against their environment [2, 3]. UAD locates agents’ *boundaries*. We ask what *drives* them, and specifically what plays the role of a discovered agent’s values. We deliberately say value *structure* rather than value *function*: locating the component that plays the value role does not tell us what that component does or prefers, and we have not built tools for that.

The hypothesis: inverting the classic AI drives argument. Omohundro’s *Basic AI Drives* [5] argues that an advanced system’s utility function “will be precious” to it, so the system will act to make that function maximally influential and maximally protected from change. In the vocabulary of Abel et al. [1], it would seek to make them maximally *empowered* and minimally *plastic*. Omohundro’s direction of explanation runs from values to structure. We hypothesize the reverse: whatever component of a system is maximally empowered and minimally plastic will, by that fact, *serve as* the system’s values. This is why we speak of *value* discovery rather than *goal* discovery: we are identifying the parts that play the value role, whatever they are. In the simple systems studied here that part is a single planted goal bit; the same structural signature could in principle locate richer value structures in richer systems.

The empowerment–plasticity trade-off [1] makes the hypothesis non-trivial: a system cannot be maximally both across its boundary, so adaptive systems have reason to specialize internally: developing belief-like parts skewed towards plasticity, value-like parts towards empowerment, and decision-making parts sitting between. Our claim is that this specialization is passively detectable.

Contributions. This paper contributes four things towards testing this hypothesis.

- (1) An operationalization of our hypothesis, the *value signature*, with procedure-mirroring noise floors and controls, designed so a non-technical collaborator could audit every step in plain English.
- (2) An instrument for unsupervised value discovery: directed-information scores computed under a menu of measurement conventions and gated by those floors. Our experiments across four worlds found a block-level convention (fuse discovered agents; offer every block as a conditioning key) that identifies every planted value at a scale where all variable-level conventions fail.
- (3) A characterization of the instrument’s limits by adversarial impostors, together with an interventional yardstick that generalizes the UAD benchmark’s goal-flip operation and resolves a case (involving a zero-noise twin) in which passive detection fails.
- (4) Several mechanism-level findings (an XOR ‘cipher’ that hides inputs from under-conditioned measures; the self-penalizing property of procedure-mirroring floors; a ‘twin channel’ by which a disconnected value persists in the value signature through its own record).

2 The value signature

This section defines the two directed measures the instrument computes, the noise floors that every score must beat, and the value signature itself.

2.1 Directed measures

The idea is to measure, for each candidate component, how strongly it pushes on everything else and how strongly everything else pushes back on it. Concretely, let X be a candidate variable in a multivariate discrete time series and R ‘the rest’ under some convention. Using the smoothed plug-in conditional mutual information of the UAD implementation (Laplace $\alpha = 0.1$, nats), we define lag-1 directed flows of transfer-entropy type [7]:

$$\text{intake (push-in): } I_{\text{in}}(X) = I(X_{t+1}; R_t | X_t), \quad (1)$$

$$\text{drive (push-out): } I_{\text{out}}(X) = I(R_{t+1}; X_t | R_t). \quad (2)$$

The *polarity* $(I_{\text{out}} - I_{\text{in}})/(I_{\text{out}} + I_{\text{in}}) \in [-1, 1]$ is the fraction of a variable’s traffic that is outbound; a raw difference is always reported alongside. Different conventions instantiate R differently (Section 3.1); an environment-flavoured drive toward a designated environment variable is also computed, never conditioned through the agent’s action, because the value’s path to the world runs through the action and conditioning on a mediator screens off precisely the flow of interest (verified analytically in Section 4).

2.2 Floors

Every score is compared to its own null distribution: the candidate’s column is circularly shifted against the fixed remainder (≥ 200 shifts, pooled across seeds where registered), and the *identical* scoring procedure is re-run, including any maximization over conditioning sets, so that search-induced bias is priced into the floor. A score counts as *above floor* if it exceeds the 95th null percentile *and* clears the null mean by $z \geq 3$ null standard deviations (the margin gate was added after a near-frozen variable was found to pass hair-thin residuals; see Section 5). We write $\text{net}(\cdot) = \max(0, \text{score} - \text{floor}_{95})$.

2.3 The value signature

A variable X exhibits the value signature if (i) its drive is above floor, (ii) its total flow $I_{\text{in}} + I_{\text{out}}$ is above floor (it is *rankable*), and (iii) $\text{net}(I_{\text{out}}) > 0$ and exceeds a minimum ratio of $\text{net}(I_{\text{in}})$ (our experiments initially used $9 \text{ net}(I_{\text{in}})$ but further work indicates a higher value performs better, see footnote 1 on page 8).

Clause (iii) is deliberately leak-tolerant: real components are all affected by their environment, so a value must not be disqualified for measurable intake, only for failing to be overwhelmingly a driver (a $9\times$ ratio corresponds to polarity ≥ 0.8 and sits well above the $\approx 3\text{--}4\times$ observed for genuine mediators). ‘Intake at floor’ is the special case $\text{net}(I_{\text{in}}) = 0$.

3 Experimental design

This section describes the worlds the experiments run in, the conditions and controls attached to every run, and the menu of measurement conventions under test.

Table 1: Worlds. All ring-coupled colonies have eight agents (≈ 49 variables) with a neighbour’s action nudging each agent’s patch on $\approx 10\%$ of steps.

World	Variables	Planted structure / probe
Anchor (Exp. 1)	9	one goal at 1.5%; two perfect lookalike logs
Colony (Exp. 2)	49	eight goals; ring coupling; partitioned variant for attribution
Deep-synergy (Exp. 2)	17	two agents, two beliefs each; parity vs. majority rule
Slow-meter (Exp. 2)	14	goal slowed to 0.5%; four causally inert meters
Puppet colonies (Exp. 3)	49	one goal replaced by a <i>captured</i> goal (fast / slow-triggered)
Alias colony (Exp. 3)	51	goal twins for two agents at noise $\{0, 0.5, 1, 2, 5\}\%$

Worlds. All worlds derive from the `SyntheticHandleWorld` of the UAD handle benchmark [8]: nine binary variables per agent forming the loop belief $B \rightarrow$ action $A \rightarrow$ environment $E \rightarrow$ sensor $S \rightarrow B$, with a goal G entering the policy ($A = B \oplus G$ up to noise) and flipping by its own coin at rate 0.015; plus a distractor D shadowing E , pure noise W , and in the original world two perfect ‘lookalike’ logs of the sensor and action lines. The conventions of these worlds were built by someone else before this project existed and for a different purpose, so were not tuned to flatter our measure. Table 1 lists the worlds; each of ours ships with an audit (a change-frequency chart, a mechanism table comparing observed disagreement rates against those predicted from the designed noise, with agreement to $\lesssim 0.1\%$, and a plain-language ‘story’), reviewed before any scoring.

Conditions and controls. Each world runs in three conditions: *main*; *no-core* (goals present but disconnected from the policy: nothing may exhibit the value signature); and *scramble* (every column independently time-shuffled: the count of above-floor scores must sit in the 5% chance band, which doubles as a calibration check on the floors). A contrast world from the UAD authors’ older multi-agent simulator supplies a variable literally named `goal_progress`, a causally inert progress meter, that the instrument must refuse.

Pre-registration and discipline. Success criteria (thresholds $\geq 80\%$ of 20 seeds; $\geq 90\%$ for goal-above-belief; uniqueness = no non-goal variable holding the value signature in more than 2 of 20 seeds) were locked before each sweep; every calibration amendment made before a sweep is dated in a decisions log. Two working rules shaped the analysis: when two requested measures disagree, the disagreement was investigated and mechanistically explained rather than averaged; and conventions were run in parallel as co-equal tests rather than a primary being chosen early. Ground-truth labels were used only to evaluate outcomes, never inside any measurement.

3.1 Measurement conventions

Table 2 lists the conventions (*tests*) evaluated. The variable-level conventions differ in how R is instantiated; the block-level conventions first run the UAD detector to partition variables into agents (its cluster-count dial chosen, as in the UAD authors’ own selection procedure, by how many variables land in blocks that still pass the Markov-blanket check), fuse each agent’s variables into a compressed macro-state block (the 64 most frequent joint states, with the probability mass lost to compression reported), and never fuse the environment: agent blocks respect a *discovered* coherence; the environment has none to respect. Orphaned variables (those the detector drops: characteristically the goals, whose rare flips carry almost no symmetric mutual information) remain individually scoreable.

Table 2: Measurement conventions. ‘Own-block’ conditions on the candidate’s own agent block; ‘any-block’ offers every block (and singleton) as a conditioning key.

Convention	Level	Rest / conditioning
pairwise	variable	sum of one-at-a-time flows
fused	variable	joint state of all other variables, both directions
fused + best-key	variable	fused intake; drive = best over no key and each single key
grown keys	variable	fused intake; drive = greedy key growth (cap 2)
fused-agents own-block	block	own agent block standing in the conditioning
fused-agents any-block	block	target block’s past + best single key element

4 Experiment 1: does the value signature identify a planted value?

The anchor world was scored under pairwise, fused, and fused+best-key, in the world as-is and with the lookalike columns removed, on 20 seeds. Before any scoring, the estimator was verified against analytic targets (a delayed copy reads $\ln 2$ nats; a chain $X \rightarrow Y \rightarrow Z$ shows $I(Z_{t+1}; X_t | Y_t) \approx 0$ while the unconditioned lag-2 dependence is positive, the mediation exhibit; the Marko–Massey conservation identity holds on analytic cases).

The cipher. A first pass showed the action’s intake reading ≈ 0.005 nats pairwise while a pre-planned fused cross-check read 0.31. Investigation, prompted by the disagreement rule, revealed the mechanism: XOR composition ($A = B \oplus G$) is a perfect cipher: each input is invisible to a measure conditioning on too little unless the other input (the ‘key’) is conditioned on. Measured readings matched hand-derived values to three decimals; re-timing the world did not help (tested), and a single extra conditioning variable restored the goal’s drive on the environment from 0.000 to 0.441 nats (hand-predicted 0.446). This motivated the key-based conventions and, later, the block-level ones.

Results. Table 3 summarizes the results of this experiment. Under fused and fused+best-key the goal exhibits the value signature in 100% of seeds, uniquely in $\geq 95\%$, and lies above the belief in polarity in 100%; all controls pass; the only intruder anywhere was pure noise W in one seed in twenty, the false-positive rate the floors are designed to admit. Under pairwise the action wires share the value signature in 20/20 seeds (as pre-registered) and, with honest floors, the goal’s own pairwise signal is indistinguishable from nothing: pairwise measurement cannot tell a value from a wire in a world with encrypted composition, and was retired, with its failure mechanism documented. Removing the lookalike columns changed no verdict, so raw data needs no pre-cleaning. The UAD detector run on the same trace recovers the agent’s body as one blanket-valid cluster (leakage 0.001 against tolerance 1.0) but drops the goal as an isolated singleton: boundary-finding and value-finding are complementary axes.

The goal_progress residual and the floor standard. Best-key refused goal_progress in only 70% of seeds, where the registered expectation was refusal outright (the contrast exists solely to be refused). Our first mechanism story blamed slowness: cumulative meters, we reasoned, would have autocorrelation-inflated shuffle floors, letting witness-like drive clear the floor while intake read as zero. Reading the stored records falsified that story: the thief was a *near-frozen* column (change rate ≈ 0.000) whose every score sat at its own unstable floor, plus initial transients from

Table 3: Experiment 1 (anchor, 20 seeds). Rates are fractions of seeds. Uniqueness = nothing but G holds the value signature; shown as-is / lookalike-free where the two configurations differ. C3 = goal_progress refused; n/a = not run.

Convention	G signature	unique	$G > B$	controls	C3 refused
fused	1.00	0.95 / 1.00	1.00	pass	n/a
fused + best-key	1.00	0.95	1.00	pass	0.70
pairwise	0.05	0.00	0.05	pass	1.00

an un-burned-in simulator. The repair became the floor standard used in every later experiment: the z -margin gate of Section 2 (a score must clear its null mean by three null standard deviations, so hair-thin residuals over an unstable floor cannot pass), a burn-in rule (the first 2,000 steps of every trace are discarded, so a simulator’s settling-in transients cannot masquerade as structure), and a reproducibility fix for the older simulator (its traces are cross-process reproducible only with a pinned hash seed, without which floors and scores can silently come from different worlds). A residual 30% theft by a genuinely moving meter is real Granger flow with no causation (it helps predict without causing anything): diffuse shared-history witnessing that no single conditioning key can screen. It became a benchmark requirement for the conventions of Experiment 2.

5 Experiment 2: which conventions survive more complex worlds?

Experiment 2 asked which measurement conventions survive outside the world they were built for. Two new conventions were first qualified against the full anchor criteria before entering the benchmark. *Grown keys* exposed its own limit during its justifying unit test: greedy growth cannot climb a flat plateau (in pure three-way parity every single key has zero marginal gain) while excelling on graded composition (majority rules, +43% over best-key); it passed C3 at 100%, falsifying our prediction that it would inherit best-key’s witness blindness, because *procedure-mirroring floors self-penalize search depth*: a deeper maximization must find real structure to outrun its own inflated null. The two block-level conventions also qualified (100% on every criterion including C3; the block context absorbed even the chance blips seen elsewhere).

Fig. 1 summarizes the results of this experiment: goal-identification rates across the four worlds under every convention. **Any-block identified every planted goal in every world**: the only convention that never missed. Mechanistically, ‘target block’s past plus one key element’ is two-block conditioning, which decrypts three-way composition and, crucially, does not require the goal to belong to a detected agent: since the detector orphans goals, this robustness is fitness for the terrain. Own-block, conditioning only on the candidate’s own block, has no ring for an orphaned goal (0% on the parity goal; 70% worst-goal in the coupled colony); the divergence between the two thus became our partition-quality diagnostic. Every variable-level convention hit a scale wall. Fused’s wall is combinatorial: the state count doubles per variable. Best-key and grown keys hit theirs on the intake side, which saturates outright at 49 variables (19,999 distinct joint states in 20,000 samples), so intake reads as bias for every variable. The *no-core control caught* this defect (0% clean seeds), and the locked interpretation rules converted it into a recorded infeasibility rather than a spurious result. The synergy cliff was measured on the parity goal (best-key 0%, grown 45%, any-block 100%; the majority goal 100% for all), and a pre-registered contingency (re-running colony failures with coupling removed) attributed both own-block’s shortfall (70% $\rightarrow$ 90%) and the single systematic thief (an action wire, 6/20 $\rightarrow$ 2/20) to ring interference, not scale.

Two defects of our own were found and fixed mid-programme by the same discipline: an early

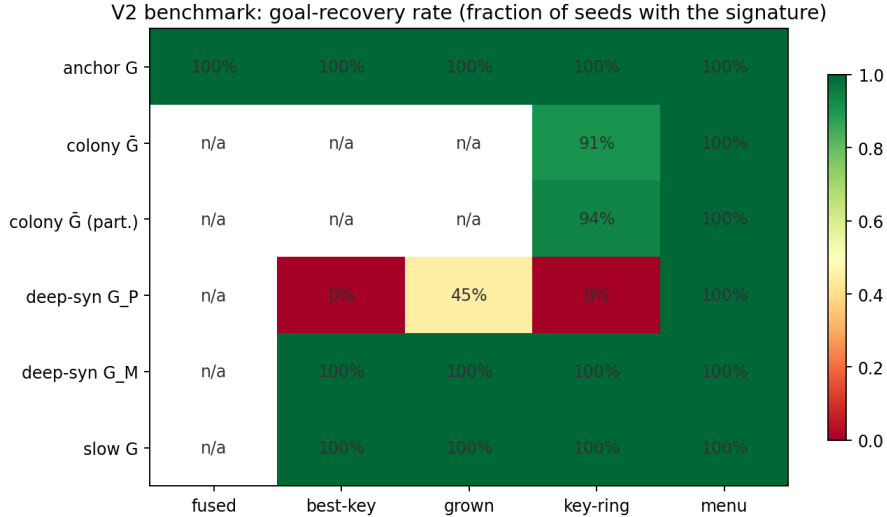


Figure 1: Experiment 2: fraction of seeds in which each planted goal exhibits the value signature, per convention (columns) and world (rows). ‘n/a’ marks recorded infeasibility (fused beyond ~ 10 variables; best-key/grown intake at colony scale). Any-block is the only column without a miss.

deep-synergy world drove both belief channels from one action, so their XOR was quasi-frozen and the parity silently collapsed (the audit had checked marginals only; it was extended to check joint dynamics), and the feasibility map had missed the intake wall.

6 Experiment 3: how easily can the instrument be fooled?

Experiments 1–2 planted values and asked whether they could be found; Experiment 3 planted impostors and asked whether they could fool the surviving conventions (any-block primary, own-block diagnostic), in colony-style worlds, with an interventional yardstick alongside.

The yardstick. The idea is to restate the value signature under interventional access: a genuine value ‘levers the world, unlevered by the world’. The UAD benchmark supplies the ingredients. It embeds a `goal_flip` operation (a *handle*, not an ideal intervention): applied to the true goal it changes the action within the tick; applied to anything else it does nothing. We generalized it to a *do-style flip* of any variable’s value as consumed by the dynamics, on 50% of interleaved steps, and built the two-direction *interventional mirror* of the value signature: a *goal-role probe* (does flipping X change its agent’s behaviour, given the agent’s policy context?) and an *autonomy probe* (does flipping X ’s own environment change X ?). The interventional verdict requires a role effect above floor *and* an autonomy effect at floor. Its construction was itself instructive: the XOR cipher bit three times (an unconditioned role probe; a saturating autonomy readout; unconditioned autonomy terms) and was caught each time by a validation bench on the anchor, where the do-style flip had to agree with the embedded operation before use; the fix, each time, was the context conditioning the benchmark’s own interventional score already employs.

Results. Table 4 summarizes the results of this experiment: for each impostor, the best passive verdict, the interventional verdict, and what interventional access adds over the best passive method, which together map the limits of passive detection. The fast captured goal (wired into the action

Table 4: Experiment 3: the limits of passive detection, per impostor, and what interventional access adds; 20 seeds; passive = any-block.

Impostor	Best passive verdict	Yardstick verdict	Access adds
Fast captured goal	refused (20/20)	refused (20/20)	none
Slow captured goal	wrongly identified (20/20)	refused (20/20)	total
Twin, 0–0.5% noise	indistinguishable (19–20/20)	twin refused (20/20)	total
Twin, $\geq 2\%$ noise	separated (19–20/20)	separated	none

but copying its own patch) was refused in 20/20 seeds: its intake is large and visible. Its audit also recorded an emergent effect: copying the patch its agent’s belief copies, it partially cancels the belief and *pacifies its own environment*. The slow captured goal (flipping at a true goal’s tempo, 1.77%, with every flip verified to follow a six-step run of its patch) **was wrongly identified by the value signature in 20/20 seeds, under both passive tests**. Its environment-intake was above its noise floor at lag 1 (net ≈ 0.03 nats), but its drive-to-intake ratio ($\approx 14\times$) passed the value signature’s leak-tolerant clause (Section 2), which was adopted so that real, leaky value cores are not disqualified. The vulnerability therefore belongs to the leak-tolerance choice, not to passive observation itself: a strictly gated passive stability test refuses this impostor at lag 1 (Experiment 5, Section 8).¹ Within the value-signature instrument itself, multi-horizon conditioning remains the evidenced remedy (the intake also surfaces above the lag-1 floor at lag 2 in 20/20 seeds). The yardstick unmasked it in 20/20 (role yes, autonomy yes). Fig. 2 gives the passive breaking point for the goal role. Twins are indistinguishable from their goals at 0–0.5% noise, as they must be: identical sequences cannot be separated by any passive method. They are transitional at 1% and separated at $\geq 2\%$ (19–20/20), with no inversions. The yardstick, by contrast, refuses the bit-identical twin even at 0% (a recording levers nothing) while its goal keeps the verdict. Interventional access resolves both cases in which passive detection fails; only the zero-noise twin is, after the comparison of Experiment 5, known to *require* it.

The twin channel. In the alias no-core condition the *disconnected* goals still exhibited the value signature in 20/20 seeds, and correctly so: a goal stripped of its behavioural wiring still causally drives its own recording, so ‘drives without being driven’ genuinely holds through the record. The instrument was right and the control’s expectation was mis-specified for twin-bearing worlds. Philosophically the point is sharp: a value can persist in the value signature through its records alone.

7 Discussion

What was established. The value signature locates a planted value structure from passive observation across single- and multi-agent worlds, and refuses decoys, disconnected goals, scrambles, and (with the right conditioning) meters; the block-level any-block convention is the only tested

¹Our pre-registered reading was different: we expected this misidentification on the ground that one step of conditioning cannot see a trigger living in run history, and at the time took the result to show exactly that. The comparison of Experiment 5 falsified that reading: the intake leak was measurable at lag 1 all along, and what admitted the impostor was the leak-tolerance clause, not passivity. Re-evaluating the stored runs at higher thresholds confirms this: the impostor’s drive-to-intake ratios span 10.3–15.0 across seeds, so thresholds modestly above this range refuse it in every seed, while the lowest ratio any true goal exhibits in any stored run is 125; the exact best threshold has not been determined experimentally.

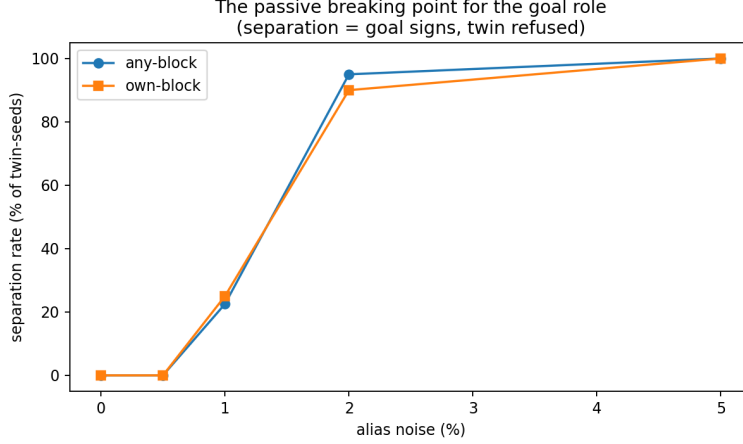


Figure 2: Experiment 3: separation rate (the goal exhibits the value signature, the twin does not) versus twin noise: the passive breaking point for the goal role lies between 1% and 2%.

instrument whole at the scale of the UAD authors’ multi-agent simulator. Its limits are now measured rather than conjectured: a history-triggered captured goal is wrongly identified, a failure that traces to the value signature’s leak-tolerant intake clause; twins below $\approx 1\text{--}2\%$ noise are passively indistinguishable. Interventional access closes both gaps, with the twin gap admitting no passive closure, giving a quantitative form of the access-model thesis of Zarncke [8]. Several findings concern method rather than result: the XOR cipher; the self-penalizing floors; the collapse-detector role of the no-core control; and the value of running disagreeing conventions in parallel rather than picking a primary early: twice, the convention we would have discarded first turned out to carry the mechanism.

Why structure, not behaviour. Behavioural inference and structural discovery are complementary, but structure has a distinctive advantage under adversarial conditions: an agent that conceals its objectives by adding decision noise weakens the behavioural signal, yet the component that most drives behaviour beneath that noise remains structurally locatable; once located, it can be isolated and studied in a regime where the noise no longer obscures it. Our results are a first indication that such structural location is possible.

Limitations. (1) *Single-variable values.* The instrument scores individual variables. It has been tested on agents with multiple belief states but not with multiple goal or value states; a value structure distributed over several components is beyond it as it stands. The grown-keys results suggest one route (deciphering interactions among several value-like variables); we conjecture a better one: use UAD’s boundary-detection machinery to identify sub-systems within an agent, then evaluate the empowerment and plasticity of those sub-systems rather than of single variables. (2) *Lag-1 conditioning together with the leak-tolerant intake clause,* which the slow captured goal jointly exploits; multi-horizon conditioning and a stricter stability gate are both evidenced remedies (the latter from the comparison of Experiment 5); what strictness costs on genuinely leaky value cores is untested. (3) *Planted structure.* All worlds are engineered; whether the polarization *emerges* in evolved or learned systems is the question the instrument now makes askable, not one it has answered. (4) The protocol was soft-blind: the implementing system had read the world’s source; protection came from the world’s independent origin, pre-registration, controls, and multi-seed

sweeps. (5) The relation between this signature and Omohundro’s drives is a hypothesis supported by recoverability, not a demonstration that empowered-and-stable components *function* as values in richer agents.

8 Further work

Two further experiments were conducted after the work reported above and inform some corrections woven into this text, as well as suggesting possible lines for future inquiry.

8.1 Experiment 4: an event-log deployment world

Experiment 4 used the instrument in a different kind of world: an event-log simulation of a deployment pipeline, developed by Gunnar Zarncke, containing agents performing different tasks and carrying an injected self-preserving utility. The experiment showed that it was possible to deploy the instrument in such an environment. The instrument identified the value structure footprint not in the outcome state the utility rewards (which was strongly driven by the agents’ environment) but in the drive-without-intake of the actions the utility selects for (regime-graded, and visible in the auditor’s log alone); in event-log worlds the mark of a value is on the choices, not the scoreboard.²

8.2 Experiment 5: an independently derived formalism compared

Our final experiment ran our instrument side by side with an interventionist, role-sufficiency-based alternative independently derived by Jobst Heitzig (roles as minimal internal representations sufficient for named external targets).³ The comparison converged wherever the formalisms overlap (including an interventional battery identical to our yardstick) and diverged instructively twice, in mirror image: a strict stability gate (theirs) passively unmasked the slow captured goal our leak-tolerant clause admits, while an evidence-only stability quantifier (theirs) let the witness class steal the goal verdict our all-source intake refuses. A new world with the value structure *outside* the agent it drives was identified by the value signature in 20/20 seeds while remaining invisible to any internal-only candidate scope; and on observation-only data the sufficiency tests cannot be stated at all, while the passive value signature runs. On this evidence, the load-bearing difference between the two routes to the same construct is the active-vs-passive contrast itself. It decides what needs a do-operator, what passivity recovers, and what can even be asked of data that only watches.

8.3 Possible lines for future inquiry

We identified the following as potentially promising lines of further experimentation.

- *Systems-of-agents environments.* Worlds in which a value structure is held by a coalition rather than a single agent. The any-block convention makes no assumption about which agent owns a value (any block can serve as the key that reveals any candidate’s drive), so the instrument is structurally ready; what is missing are environments that actually plant coalition-held values to identify.
- *Multi-horizon conditioning.* The slow captured goal’s trigger surfaces above the lag-1 floor at lag 2 in 20/20 seeds, so extending the conditioning window beyond one step is an evidenced

²<https://github.com/SJ-Beard/deployment-pipeline-value-detect>

³<https://github.com/SJ-Beard/active-v-passive-value-detect>

upgrade rather than a speculation; the design question it opens is how to keep the noise floors procedure-mirrored and honest as the window (and with it the state space) grows.

- *Sub-system-level scoring.* Score discovered sub-systems rather than single variables, using the UAD boundary machinery *inside* an agent to propose components and then evaluating each component’s empowerment and plasticity, to reach value structures distributed over several variables, which the present instrument cannot represent.
- *The twin channel as a persistence probe.* A goal stripped of its behavioural wiring still drives its own recording, so the value signature survives through records (found in Experiment 3’s no-core control). Deliberately exploiting this would probe how long, and through what kinds of record, a value structure remains locatable after it stops steering behaviour.
- *A synthesis instrument.* Combine the value signature’s world-directed, all-source intake with the strict stability gate and an explicit uniqueness pressure; the reversals of Experiment 5 define its requirements, and the open cost question is what strictness loses on genuinely leaky value cores, which the leak-tolerant clause was designed to protect.

Statement on use of AI

The author conceived the hypothesis and framing, made all design decisions, reviewed every audit and pre-registration in plain language, and directs the programme. Claude (Anthropic), an AI system, wrote all code, ran the experiments and analyses, and drafted this manuscript under the author’s direction; every decision, deviation, and defect is dated in the project’s decisions log. **This manuscript is, as it stands, AI-generated:** it was written by Claude and reviewed and approved by the author, who intends to write the account in their own words when time allows. In line with the standing convention that AI systems are not listed as authors (authorship carries a responsibility that cannot be delegated to them), the author takes full responsibility for the content. The manuscript is not being submitted for publication.

Reproducibility

All code (`value_detect`), world generators, audits, locked pre-registrations, verdict tables, and figures are archived with fixed seeds; the UAD benchmark code is imported unmodified. Sweeps ran on a single laptop (Experiment 1 $\approx$ 1 h; Experiment 2 $\approx$ 12.5 h; Experiment 3 $\approx$ 37.5 h at four workers).

References

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